

### Question #1 of 49

Question ID: 1590532

Compared to the Sharpe ratio, the Sortino ratio may be preferred when comparing hedge funds due to which of the following?

- A) The high volatility of hedge funds.
- B) The right-tail risk of hedge funds.
- C) The left-tail risk of hedge funds.



#### Explanation

Due to the left-tail risk of hedge funds, the Sortino ratio may be superior in defining risk since it only considers downside deviation. Both measures would reflect high volatility through their risk measures.

(Module 34.4, LOS 34.i)

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### Question #2 of 49

Question ID: 1590530

A process which is helpful for creating linear conditional factor models that avoid multicollinearity is:

- A) step-wise regression analysis.
- B) correlation residual factor analysis.
- C) mean-variance collinearity analysis.



#### Explanation

Step-wise regression was run by Hasanhodzic and Lo in their 2007 paper and demonstrated that removing the variables Bond and Commodity from their model resulted in a higher  $R^2$  than prior analysis which included these variables. They concluded there was multicollinearity and removing those factors improved the model.

(Module 34.4, LOS 34.h)

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### Question #3 of 49

Question ID: 1590503

Managed futures strategies are typically characterized as:

- A) a strategy with a return profile that tends to be very counter-cyclical. 
- B) a highly liquid strategy, active across a wide range of asset classes, and able to go long or short with relative ease. 
- C) an diversifying strategy to consider in rising markets as they outperform equity markets in rising markets. 

### Explanation

Managed futures strategies are typically characterized as highly liquid, active across a wide range of asset classes, and able to go long or short with relative ease. High liquidity results from futures markets being among the most actively traded markets in the world. Futures contracts also provide highly liquid exposures to a wide range of asset classes that can be traded across the globe 24 hours a day. Because futures contracts require relatively little collateral to take positions as a result of the exchanges' central clearinghouse management of margin and risk, it is easier to take long and short positions with higher leverage than traditional instruments.

The returns of managed futures tends to be very cyclical. Between 2011 and 2018, the directionality of foreign exchange and fixed-income markets deteriorated, volatility levels in many markets dissipated, and periods of acute market stress temporarily disappeared. Except for equity markets in some developed countries, many markets became range-bound or mean-reverting, which hurt managed futures. The diversifying benefit of trend-following strong equity markets is also less diversifying to traditional portfolios than if such trends existed in other non-equity markets.

The value added from managed futures has typically been demonstrated during periods of market stress; for example, in 2007–2009 managers using this strategy benefitted from short positions in equity futures and long positions in fixed-income futures at a time when equity indexes were falling and fixed-income indexes were rising.

(Module 34.3, LOS 34.e)

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### Question #4 of 49

Question ID: 1590511

Hedge Fund T is a hedge fund that employs market neutral, opportunistic, and specialist strategies as part of its operations. A hospital endowment fund hires Hedge Fund T to manage part of its portfolio. This situation is *best* described as:

- A) a global macro fund. 
- B) a multistrategy fund. 
- C) a fund-of-fund. 

### Explanation

Since the hedge fund is operating several strategies, the best description is a multistrategy fund. The fund-of-funds would best relate to different hedge funds each employing a different strategy. A global macro is a type of opportunistic strategy.

(Module 34.3, LOS 34.g)

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### Question #5 of 49

Question ID: 1590490

Which of the following *best* represents the net short position of a short-biased hedge fund strategy?

- A) 100%. 
- B) 0%. 
- C) 50%. 

#### Explanation

The short-biased strategy does have some long exposure, leaving the net short exposure typically 30%–60% of the portfolio. The 100% net short would be indicative of a dedicated short strategy and a market neutral strategy may have a net short of 0%.

(Module 34.2, LOS 34.b)

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### Question #6 of 49

Question ID: 1590486

The failure of a merger to occur is a risk of which of the following hedge fund strategies?

- A) Opportunistic. 
- B) Equity related. 
- C) Event driven. 

#### Explanation

Since the success of an event-driven strategy is dependent upon the event occurring, the failure of a merger to occur is a risk of that type of strategy. Equity-related strategies focus on stocks, and hence the primary source of risk is equity risk. Opportunistic strategies employ a top-down approach, often consider multiple asset classes, and vary with market conditions.

(Module 34.1, LOS 34.a)

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### Question #7 of 49

Question ID: 1590512

Which of the following is a disadvantage of a multistrategy fund as opposed to a fund-of-funds?

A) Volatility of returns.



B) Fee structure.



C) Operational risk.



#### Explanation

Since the funds in a multistrategy fund are operated by the same firm, any operational flaws would not be diversified away as with a fund-of-funds. Fees are likely more attractive for a multistrategy fund due to netting, and both types of funds would seek to limit volatility.

(Module 34.3, LOS 34.g)

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### Question #8 of 49

Question ID: 1590488

Which general statement describing key differences between long/short equity (L/S) strategies and equity market neutral (EMN) is *most accurate*?

L/S strategies typically have more concentrated portfolios than EMN strategies,

A) use lower leverage than EMN, and target unlevered returns that are higher than EMN.



EMN strategies rely on selecting securities with sufficient liquidity, have longer

B) security selection time horizons, and usually have greater leverage than L/S equity strategies.



C) EMN strategies are less diversified than L/S strategies but typically use greater amounts of financial leverage to generate returns than L/S strategies.



#### Explanation

L/S strategies are typically more fundamental in nature as they select specific securities to go long and short. L/S funds attempt to earn alpha on both sides of the trade and target a lower beta while achieving returns consistent with long-only equities. EMN strategies has systematic approaches in selecting long and short candidates, trade more frequently, and are close to market neutral. In order to deliver higher returns, these funds use substantial leverage.

(Module 34.2, LOS 34.b)

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### Question #9 of 49

Question ID: 1590504

Catalysts for global macro hedge funds outperformance include:

- A) muted volatility in periods of quantitative easing and p/e multiples can expand across the globe. 
- B) steep equity market sell-offs, interest rate changes, currency devaluations, volatility spikes and geopolitical shocks. 
- C) low and stable interest rates, steady trending equity global markets and tight credit spreads. 

#### Explanation

Global macro hedge fund managers take investment positions based on their future predictions of various global economic variables including inflation, currency exchange rates, yield curves, central bank policies, and the general economic health of different countries. These fund managers anticipate changes before other market participants and then invest in a position and wait for the rest of the market to come around. Thus, this strategy has aspects of a contrarian nature with some managers outperforming during times of market stress.

(Module 34.3, LOS 34.e)

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### Question #10 of 49

Question ID: 1590485

Which one of the following is *least likely* considered to be an argument to include hedge funds in a diversified portfolio?

- A) Access to a larger investment universe. 
- B) Greater transparency of investment management decisions and ability to pursue more aggressive strategies. 
- C) Flexibility to use short selling and derivative contracts and greater use of leverage. 

#### Explanation

Greater transparency is generally not an attributed to hedge funds as many managers attempt to hide their activities for competitive purposes.

(Module 34.1, LOS 34.a)

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### Question #11 of 49

Question ID: 1590492

The hedge fund strategy that typically requires the longest time commitment and has the greatest variability of investment returns is:

- A) distressed securities. 
- B) fixed income arbitrage. 
- C) merger and acquisition funds. 

#### Explanation

Relative to other hedge fund strategies, distressed debt takes a long time to work out as legal proceedings grind through the system. Distressed debt can generate spectacular investment returns or complete losses.

(Module 34.2, LOS 34.c)

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### Question #12 of 49

Question ID: 1590491

The payoff profile of merger arbitrage positions is similar to the return on:

- A) a risky bond, short a call and long a put. 
- B) a riskless bond, short a put, and long a call. 
- C) a riskless bond, short a call, and short a put. 

#### Explanation

The strategy incorporates the time associated between merger announcement and transaction closing which is discounted at the risk free rate. M/A hedge funds also earn additional "premium" from equity holders which reflects the potential that the deal may not successfully close. Similarly, hedge funds are in effect, long a call if a higher bid were to surface. All cash flows are discounted at the risk free rate.

(Module 34.2, LOS 34.c)

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### Question #13 of 49

Question ID: 1590496

Because convertible securities are issued sporadically by smaller companies in small offering sizes with unrated debt, the value of the embedded option tends to trade at:

- A) relatively low implied volatility levels compared with realized volatility for the underlying equity. 

**B)** a discount to the implied equity volatility of the underlying equity.



**C)** a premium to implied volatility of the underlying equity.



### Explanation

Negative issues for investors like small issue size, poor covenant protection, lack of institutional familiarity with the issue and no-rated credits result in contraction in overall valuation which represents itself as lower implied volatility compared with historic observations. These factors contribute to the convertible option being undervalued.

(Module 34.2, LOS 34.d)

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A hedge fund analyst working for a corporate pension fund uses a conditional linear factor risk model to identify the sources of return for the hedge fund Uno Investments (UNO).

Through stepwise regression, the model identifies four independent factors:

1. Equity risk (SNP500)
2. Currency risk (USD)
3. Credit risk (CREDIT)
4. Volatility risk (VIX)

Using monthly returns for the last 10 years, the coefficient estimates from the model and corresponding *t*-statistics are displayed in **Exhibit 1: Conditional Linear Factor Model Coefficients for UNO**.

### Exhibit 1: Conditional Linear Factor Model Coefficients for UNO

| UNO         |          |                     |
|-------------|----------|---------------------|
| Coefficient | Estimate | <i>t</i> -Statistic |

#### Normal Times

|        |        |       |
|--------|--------|-------|
| USD    | 0.095  | 0.65  |
| CREDIT | 0.015  | 0.12  |
| SNP500 | 0.678  | 7.42  |
| VIX    | -0.183 | -2.67 |

### Crisis Times (Incremental)

|         |        |       |
|---------|--------|-------|
| DUSD    | 0.327  | 1.49  |
| DCREDIT | -0.251 | -2.20 |
| DSNP500 | -0.189 | -2.35 |
| DVIX    | +0.054 | +2.88 |

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UNO is an equity long/short fund with the following statement in its fund documents regarding its strategy:

UNO is a sector-focused long/short manager that aims to identify attractive alpha-generating opportunities on both the long and short side in the biotechnology sector. The fund uses a two-pronged approach: a team of research analysts work with the fund manager to identify the best bottom-up investments in the biotechnology sector, while macro-focused analysts work with the fund manager to apply overlay strategies using futures and options to adjust the market exposure of the fund.

The analyst uses the model to assess global macro hedge fund managers in their investment universe for superior market timing skills. They base their assessment on the signs and statistical significance of the coefficients of the model.

The investment committee of the corporate pension fund has asked the investment management team to review the rationale for including an allocation to hedge funds in the pension portfolio.

### Question #14 - 17 of 49

Question ID: 1590535

Based on the data in **Exhibit 1: Conditional Linear Factor Model Coefficients for UNO** and the statement from the fund documents regarding UNO's strategy (and assuming that a t-statistic with an absolute value greater than 2 is significant), which of the following statements is *most accurate*?

- A) The macro analysts have exhibited good market timing skill with respect to equity risk.





- B) UNO has been, on average, net short equity market risk during times of market crisis. 
- C) The macro analysts have exhibited poor market timing skill with respect to equity risk. 

### Explanation

The output of the model shows that UNO has significant long exposure to equity risk during normal market conditions (with a statistically significant coefficient of 0.678). The coefficient of DSNP500 being equal to  $-0.189$  indicates that the manager is less long in market crises (an overall exposure of  $0.678 - 0.189 = 0.489$ ). So, there is skill by the macro analysts in reducing market risk during a crisis. The data shows market timing skill in reducing the equity market exposure during a crisis, and the data does not imply that the manager has a negative exposure to market risk during crises (as stated previously, this coefficient remains positive at 0.489).

(Module 34.4, LOS 34.h)

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### Question #15 - 17 of 49

Question ID: 1590536

Based on the data in **Exhibit 1: Conditional Linear Factor Model Coefficients for UNO**, which of the following statements regarding exposure of UNO to the volatility factor is *most accurate*?

- A) The manager has destroyed value through poor management of volatility exposure in crisis times versus normal times. 
- B) The manager has negative exposure to volatility in times of market crisis. 
- C) The macro analysts may be controlling for market exposure through long put positions on equity markets. 

### Explanation

The data shows that in normal times, the manager is short volatility (with a significant VIX coefficient of  $-0.183$ ) and that in times of crisis, the volatility exposure becomes less negative (taking a value of  $-0.183 + 0.054 = -0.129$ ). The manager has negative exposure to volatility in times of market crisis; hence, this is the correct answer. Reducing the negative exposure to volatility in times of crisis when volatility is rising will create value rather than destroy value in the fund. Long put positions will have a positive volatility exposure, which is not evidenced by the negative volatility exposures of the model coefficients.

(Module 34.4, LOS 34.h)

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### Question #16 - 17 of 49

Question ID: 1590537

A manager with superior market timing skills would exhibit which of the following statistically significant coefficients in the conditional linear factor model?

**A)** USD: negative, DUSD: positive, CREDIT: positive, DCREDIT: negative.



**B)** USD: positive, DUSD: negative, CREDIT: positive, DCREDIT: negative.



**C)** USD: negative, DUSD: negative, CREDIT: positive, DCREDIT: positive.



#### Explanation

A manager with superior market timing skills in trading the U.S. dollar will be short USD in normal times, when riskier assets are outperforming; they will increase their USD position in times of crisis, when the USD is appreciating.

A manager with superior market timing skills in trading credit will be long credit risk in normal times, when spreads are flat/narrowing; they will lower their credit exposure in times of crisis, when spreads are widening.

(Module 34.4, LOS 34.h)

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### Question #17 - 17 of 49

Question ID: 1590538

Which of the following rationales is *least likely* to be appropriate when justifying a 20% allocation to hedge funds in a multi-asset portfolio such as a corporate pension plan?

**A)** The allocation will decrease portfolio volatility.



**B)** The allocation will increase portfolio risk-adjusted return.



**C)** The allocation will increase portfolio return.



#### Explanation

Adding a 20% allocation to a hedge fund strategy in a traditional portfolio typically leads to the portfolio having lower volatility, higher risk-adjusted return as measured as a Sharpe ratio or Sortino ratio, and lower maximum drawdown. It does not typically increase the overall return of the portfolio.

(Module 34.4, LOS 34.i)

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Leigh Winstanton is a relative value hedge fund manager. She is currently analyzing government bond and swaps markets for pricing discrepancies, collating the data displayed as follows:

- 5-year Treasury Inflation-Protected Securities (TIPS) coupon rate: 1% (priced at par)
- 5-year Treasury bond coupon rate: 3% (priced at par)
- 5-year inflation swap fixed rate: 1.5%

Winstanton also engages in convertible bond arbitrage trades. She collates data on a potential convertible arbitrage trade in the securities of Triste, Inc. (TST), a medium-sized manufacturer of agricultural equipment, displayed as follows:

- TST has in issue a 2-year 4% annual coupon convertible bond, priced at 115
- The conversion ratio of the TST convertible bond is 25 shares per USD 1,000 par
- Current price of TST shares is USD 50 per share
- Borrowing costs are USD 0.60 per year, per share
- Dividend per share is expected to be USD 1 per year, per share

Winstanton is investigating whether there is an immediate arbitrage opportunity from buying the convertible bond and converting into shares. Winstanton is also interested in how carrying this convertible bond arbitrage position through time will likely affect the profitability of the trade.

Winstanton is looking to hire a volatility trader to execute trades with the objective of hedging the exposure of the fund's existing relative value positions to large, unexpected movements in spreads and prices in times of market stress.

### Question #18 - 21 of 49

Question ID: 1590519

To capture the pricing discrepancy between the Treasury bond market and the swaps market, Winstanton should execute which of the following trades?

- A)** Buy Treasuries, Sell TIPS, pay fixed under inflation swap.
- B)** Sell Treasuries, sell TIPS, receive fixed under inflation swap.
- C)** Buy Treasuries, buy TIPS, pay fixed under inflation swap.



#### Explanation

The manager can earn excess return as follows:

- Buy Treasuries, locking in a fixed inflow coupon of 3%
- Short sell TIPS, locking in coupon outflow of 1% + future realized inflation
- Pay fixed leg of 1.5% and receive future realized inflation under an inflation swap

Net exposure = 3% – (1% – inflation) – (1.5% – inflation) = 0.5%

(Module 34.3, LOS 34.e)

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### Question #19 - 21 of 49

Question ID: 1590520

The immediate arbitrage profit from buying the TST convertible bond and converting into shares is *closest* to:

- A) USD 6 per share.
- B) USD 4 per share.
- C) USD 10 per share.



#### Explanation

Purchasing USD 1,000 par of the bond at a price of 115 per cent of par costs  $\text{USD } 1,000 \times (115 / 100) = \text{USD } 1,150$ .

The cost per share of owning the shares through buying the convertible bond and converting is, therefore,  $\text{USD } 1,150 / 25 = \text{USD } 46$ .

The current shares are worth USD 50 in equity markets; hence, investors can lock in a USD 4 gain immediately from buying the convertible and short selling the shares today.

(Module 34.2, LOS 34.d)

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#### Question #20 - 21 of 49

Question ID: 1590521

Introducing the costs and benefits of carrying the convertible arbitrage trade through time will:

- A) increase the profitability of the trade.
- B) decrease the profitability of the trade.
- C) have no impact on the profitability of the trade.



#### Explanation

Annual coupon income from the long convertible bond position will be  $\text{USD } 1,000 \times 4\% = \text{USD } 40$  per USD 1,000 par.

The annual cost of borrowing and dividend that needs to be paid to the stock lender equals  $\text{USD } 0.60 + \text{USD } 1 = \text{USD } 1.60$  per share. In total, this will cost  $\text{USD } 1.60 \times 25 = \text{USD } 40$  per USD 1,000 of par.

The coupon income is equal to the outflows of costs of borrowing and dividends; hence, carrying the position through time will not affect the overall profitability of the trade.

(Module 34.2, LOS 34.d)

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#### Question #21 - 21 of 49

Question ID: 1590522

Considering the objective of the new volatility trader, which of the following volatility trading strategies would *most likely* be appropriate for adding to the fund?

- A) Long positions in OTC options. 
- B) Roll down using VIX futures. 
- C) Relative value volatility arbitrage using exchange-traded options. 

#### Explanation

In times of market stress, volatility increases—hence, to meet the objective of providing protection in times of market stress, the volatility trader should execute a long volatility strategy. The only strategy listed that is a long volatility strategy is taking long positions in OTC options. "Rolldown" profits are earned by selling long-dated VIX futures when the term structure of volatility is positively sloped, and benefiting from falling prices as futures fall over time. As such, the rolldown strategy is a short volatility strategy, not a long volatility strategy—so, it is inappropriate. A relative value volatility arbitrage strategy involves buying cheap implied volatility and selling expensive implied volatility—hence, it is volatility neutral rather than long volatility in nature.

(Module 34.3, LOS 34.f)

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#### Question #22 of 49

Question ID: 1590494

Institutional limitations on banks and insurance companies are *most likely* to lead to pricing inefficiencies in which type of strategy?

- A) Volatility trading. 
- B) Distressed securities. 
- C) Global macro. 

#### Explanation

Limits to hold-only investment-grade debt by banks and insurance companies may lead to pricing inefficiencies for distressed securities, which can be capitalized upon. While there may be other limitations concerning macro and volatility strategies, the questions specifically asks about pricing inefficiencies.

(Module 34.2, LOS 34.c)

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#### Question #23 of 49

Question ID: 1590531

Including which of the following hedge fund strategies in a diversified portfolio is *least likely* to mitigate drawdowns?

A) Global macro.



B) Distressed securities.



C) Equity market-neutral.



#### Explanation

An explanation for the inability of distressed securities to mitigate drawdowns is the long time horizon of distressed debt investments and their direct exposure to both equity and credit risk.

(Module 34.4, LOS 34.i)

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### Question #24 of 49

Question ID: 1590487

Opportunistic hedge fund strategies employ:

A) long and short positions in equity.



B) an investment in distressed debt.



C) a top-down approach.



#### Explanation

Opportunistic strategies have a macro focus and generally follow a top-down approach. The other answers are more closely associated with a bottom-up approach.

(Module 34.1, LOS 34.a)

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### Question #25 of 49

Question ID: 1590528

A conditional factor risk model is used in analyzing hedge fund returns. The purpose of the dummy variable in that model is to distinguish between:

A) periods of normal market activity and financial crises.



B) funds in countries with high inflation and low inflation.



C) managed funds and passive funds.



#### Explanation

The dummy variable is employed in the model since factors may have different influence during normal periods and financial crises. Such a model is termed a *conditional linear factor model*, where the dummy variable allows for the conditional analysis.

(Module 34.4, LOS 34.h)

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**Question #26 of 49**

Question ID: 1590510

Two key advantage of multi-strategy hedge funds over fund-of-funds managers are:

- A) economies of scale as different hedge fund strategies share the same administrative costs, and greater left tail risk than fund-of-funds. 
- B) the ability and knowledge to rapidly make tactical reallocation decisions to the firm's strategic allocation based on changing market conditions, and greater visibility to assess the overall risk exposure. 
- C) more attractive fees for investors than fund-of-funds managers, and a greater level of diversification than fund-of-funds provide. 

**Explanation**

Multi-strategy managers operate multiple strategies under one organization and can react or adjust to anticipated market conditions more rapidly than fund-of-funds managers, which frequently have lockups and gates to negotiate when making allocation changes.

(Module 34.3, LOS 34.g)

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**Question #27 of 49**

Question ID: 1590506

Which of the following is a characteristic of a managed futures strategy as a result of crowding?

- A) Price inefficiency. 
- B) High leverage. 
- C) Execution slippage. 

**Explanation**

Execution slippage is the result of crowding as participants pursue similar strategies. While high leverage is a feature of managed futures, it is not specifically related to crowding. While exploiting pricing inefficiencies are often an objective of hedge funds, that characteristic is not specifically related to the question either.

(Module 34.3, LOS 34.e)

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**Question #28 of 49**

Question ID: 1590533

A traditional 60% equity/40% fixed income allocation was adjusted by adding a hedge fund to the portfolio, resulting in the following allocation: 48% equity/32% fixed income/20% hedge fund. There was no reduction in the standard deviation of the portfolio after the addition of the hedge fund. The hedge fund *most likely* added was:

- A) a dedicated short bias fund.
- B) a distressed securities fund.
- C) a bear market equity fund.



#### Explanation

Distressed securities investing involves long positions that are frequently large successes or failures, thus limiting their ability to reduce standard deviation. The other strategies reflect less exposure to long positions and would likely reduce standard deviation.

(Module 34.4, LOS 34.i)

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#### Question #29 of 49

Question ID: 1590529

Which of the following is *least likely* included in the four-factor, conditional linear model used to quantify hedge fund strategies' risk exposures?

- A) Credit risk.
- B) Volatility risk.
- C) Interest rate risk.



#### Explanation

The four risk factors are credit risk, volatility risk, equity risk, and currency risk.

(Module 34.4, LOS 34.h)

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#### Question #30 of 49

Question ID: 1590508

Hedge Fund Z follows a short volatility strategy, while Hedge Fund Y makes insurance investments. Which of the following is *most accurate* concerning the funds?

- A) Z has a natural hedge against market declines.
- B) Z's strategy is neither positively nor negatively correlated with the market.
- C) Y's investments will be largely uncorrelated with market movements.





## Explanation

A primary benefit of insurance strategies is the lack of correlation with the market. A short volatility strategy would have poor returns as volatility increases. Since markets tend to fall during volatile periods, it does not provide a hedge. Similarly, the volatility strategy will have a positive correlation with the market.

(Module 34.3, LOS 34.f)

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Christine Kelly is chief investment officer of Pontoon Asset Management (PAM), a large multi-strategy hedge fund. She is meeting with all the management teams at PAM to understand their recent performance and current market positioning.

Kelly first meets with Nicola Shore, head of the equity long/short team. Shore describes a recent pairs trade executed in two retailers: Homestore (HOM) and Sarks (SAR). Shore explains that the team took a long position in HOM and a beta-neutral short position in SAR with the expectation that HOM will execute a better business model, with respect to the migration from physical to online retailing. Kelly notes that HOM has a stock market beta of 1.1 and a stock price of USD 40, and SAR has a stock market beta of 0.55 and a stock price of USD 80.

Kelly next meets with Luca Cohen, head of the merger arbitrage team. Cohen states that due to increasing competition and a reduction in opportunities in their traditional hard-catalyst investment universe, the team has begun executing more soft-catalyst trades. Kelly asks Cohen how this is likely to affect the risk and return of the investments in the future.

Kelly asks about recent trades executed by the team and Cohen presents details regarding the details of a hard catalyst trade executed in relation to a merger between Bigco (BIG) and Small, Inc. (SMA), displayed in **Exhibit 1: Details on Merger Arbitrage Trade in BIG and SMA**:

### Exhibit 1: Details on Merger Arbitrage Trade in BIG and SMA

|                                                  |                                      |
|--------------------------------------------------|--------------------------------------|
| <b>Deal Terms</b>                                | 0.5 shares of BIG for 1 share in SMA |
| <b>BIG share price (pre-) deal announcement</b>  | USD 54                               |
| <b>BIG share price (post-) deal announcement</b> | USD 52                               |
| <b>SMA share price (pre-) deal announcement</b>  | USD 23                               |
| <b>SMA share price (post-) deal announcement</b> | USD 25                               |

Finally, Kelly meets with Alan Rode, head of the distressed securities team at PAM. Kelly has the least knowledge regarding distressed strategies out of all of the hedge fund strategies and asks Rode to describe recent trades undertaken by the fund. Rode presents to Kelly a capital structure arbitrage trade in a large retail department store company.

**Question #31 - 34 of 49**

Question ID: 1590499

In order to establish a beta-neutral pairs trade, relative to the number of shares bought in the long position in HOM, Shore should short sell:

- A) the same number of shares in SAR.
- B) double the number of shares in SAR.
- C) half the number of shares in SAR.

**Explanation**

Shore should short sell the same number of shares in SAR as the number of shares held in the long HOM position to establish a beta-neutral trade.

Because SAR has a beta equal to half the beta of HOM, on a beta-risk basis alone, the pairs trade will need twice as many shares in SAR as HOM. However, because the share price of SAR is twice the price of shares in HOM, on a market value basis only half the number of shares in SAR are needed with respect to the number of shares of HOM. Therefore, creating a beta-neutral position will involve holding the same number of shares short in SAR as the number of shares bought in the long HOM position.

(Module 34.2, LOS 34.b)

**Question #32 - 34 of 49**

Question ID: 1590500

The increase in soft-catalyst trades by the merger arbitrage team will be expected to:

- A) increase the return and decrease the risk of the team.
- B) decrease both the risk and return of the team.
- C) increase both the risk and return of the team.

**Explanation**

Under a soft-catalyst approach, investments are made based on expected announcements of mergers. Under a hard-catalyst approach, investments are made based on merger announcements that have already been made. Because the soft-catalyst approach involves the extra uncertainty of deals being announced, profits from this approach are expected to be more volatile. However, because they would profit from the large share price movements when deals are announced, there would be higher expected returns.

(Module 34.2, LOS 34.c)

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### Question #33 - 34 of 49

Question ID: 1590501

The ratio of potential losses if the deal fails versus the potential gains if the deal succeeds for the BIG/SMA merger arbitrage trade is *closest* to:

A) 4.



B) 3.



C) 5.



#### Explanation

If Cohen has established a USD 10,000,000 long position in SMA, then they have bought  $\text{USD } 10,000,000 / \text{USD } 25 = 400,000$  shares in SMA post/after deal announcement.

For each one of these SMA shares, they will receive 0.5 shares of BIG if the deal goes through, which sets the size of the short position in BIG as follows:

Number of BIG shares to short =  $0.5 \times 400,000 = 200,000$ .

Short proceeds raised from short selling 200,000 shares of BIG at a price of USD 52 = USD 10,400,000. If the deal goes through the short position in BIG, it can be covered by the BIG shares received in exchange for the long position in SMA. This leaves the fund with profits of short proceeds – investment in SMA shares =  $\text{USD } 10,400,000 - \text{USD } 10,000,000 = \text{USD } 400,000$ .

Should the deal fail, then, assuming prices revert to the pre-deal announcement level, the following occur:

The loss from a long position in SMA =  $400,000 \times (\text{USD } 25 - \text{USD } 23) = \text{USD } 800,000$ .

The loss from a short position in BIG =  $200,000 \times (\text{USD } 54 - \text{USD } 52) = \text{USD } 400,000$ .

Total loss =  $\text{USD } 800,000 + \text{USD } 400,000 = \text{USD } 1,200,000$ .

Hence, the ratio of potential losses to potential gains =  $\text{USD } 1,200,000 / \text{USD } 400,000 = 3x$ .

(Module 34.2, LOS 34.c)

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Which of the following statements regarding a capital structure arbitrage trade is *most accurate*?

- A) Capital structure arbitrage trades are not expected to be profitable when the distressed company is forced into liquidation. 
- B) Capital structure arbitrage opportunities usually occur due to equity retail investors being too bearish on the fortunes of a distressed company versus the outlook of debt investors in the company. 
- C) Capital structure arbitrage trades can result in the investor taking an equity stake in new equity of the distressed company post-bankruptcy reorganization. 

#### Explanation

In a capital structure arbitrage trade, an investor takes a long position in the debt securities of a distressed company and short sells the equity. Should the company enter a bankruptcy reorganization, the investor might swap their debt for equity and become an equity investor in the new post-reorganization company.

A capital structure arbitrage opportunity usually occurs when equity retail investors are too bullish on the fortunes of the distressed company, while debt investors are pricing in a higher chance of bankruptcy—thereby making the debt cheap relative to equity.

Should the company be forced into liquidation, it is likely the short equity position will be worthless, but the long position in the corporate debt will have some liquidation recovery value; hence, the trade is profitable.

(Module 34.2, LOS 34.c)

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#### Question #35 of 49

Question ID: 1590507

The key factors an insurance settlements portfolio manager must successfully analyze include:

- A) the underlying interest rate on the policy, the early termination provisions of the contract, and the government bond yield curve. 
- B) expected policy cash flows, ongoing premium payment obligations, and the eventual death benefit to be received. 
- C) the probability that the insurance company will face financial pressure, the demeanor of the policy holder, and the age of the policyholder. 

#### Explanation

Insurance settlement managers must be able to create a discounted cash flow analysis of all future cash flows. Continuing premium payments owed, buyout terms for existing holder, and eventual death benefits to be received at an uncertain point in the future are key factors they must consider.

(Module 34.3, LOS 34.f)

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Bobby Berg is an investment analyst at Conduit Asset Management (CAM), a manager of a large fund of hedge funds. Berg is performing research into investments relating to catastrophe reinsurance and life settlements, a relatively new sector for hedge fund investment.

Berg meets with Michael Stern, a hedge fund manager who specializes in investments in the insurance sector. Stern discusses recent life settlement opportunities presented by insurance brokers and discusses their relative merits, the details of which are displayed in Exhibit 1.

**Exhibit 1: Selected Life Settlement Opportunities**

| <b>Insurance Pool</b> | <b>Surrender Value (\$m)</b> | <b>Annual Premium (Percentage of Benefit)</b> | <b>Life Expectancy of Pool vs. Mortality Tables</b> |
|-----------------------|------------------------------|-----------------------------------------------|-----------------------------------------------------|
| A                     | 10.4                         | 0.95%                                         | Longer                                              |
| B                     | 9.2                          | 0.87%                                         | Shorter                                             |
| C                     | 12.8                         | 1.25%                                         | Shorter                                             |

Berg is performing due diligence on hedge fund managers that could potentially be added to the existing fund of hedge funds portfolio. She is aware of a recent motion passed at an investment committee meeting, which recorded the following short-term tactical objectives:

- Increase liquidity
- Decrease leverage
- Lower exposure of the fund to tail risks, such as acute credit weakness in markets

Berg considers how best to achieve these objectives when considering the existing style allocation of CAM.

CAM provides the following information on its fee structure to potential clients:

- CAM charges 1% management fee and 5% incentive fee
- The average underlying fund charges 1.5% management fee and 17.5% incentive fee

A client of CAM asks Berg about the relative merits of a fund-of-funds structure versus a multi-manager hedge fund structure. They are specifically interested in which structure

would give the best result for investors, with respect to the following:

- Operational risk
- Speed of tactical asset allocation
- Fee netting risk

### Question #36 - 39 of 49

Question ID: 1590514

Using the data in Exhibit 1 and assuming that all other features of Insurance Pool A, Insurance Pool B, and Insurance Pool C are equivalent, the life settlement investment opportunity with the best expected return is *most likely* to be:

- A) Pool C. 
- B) Pool A. 
- C) Pool B. 

#### Explanation

Life settlement investments involve the hedge fund buying existing insurance policies from insured lives, paying the premium for the remainder of the contract's life, and then receiving the payout of benefits on the death of the original policyholder. Hedge funds will prefer pools of insurance contracts that have a low surrender value (defined as the value the insured lives could receive from the original insurance company if they cancel their contract) because this implies that policyholders would be happy to sell their insurance contracts for a relatively lower amount. Hedge fund managers would prefer insurance pools with a lower premium as a percentage of benefit because they will have to pay this for the remainder of the life of the original policyholder. Hedge funds would also prefer contracts relating to lives with lower life expectancies because they will receive benefit payouts sooner if the original policyholders die earlier. Pool B meets these conditions of lower surrender value, lower annual premium, and shorter life expectancy.

(Module 34.3, LOS 34.f)

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### Question #37 - 39 of 49

Question ID: 1590515

Considering all the short-term objectives specified in the recent investment committee motion, which of the following style allocation recommendations for CAM is *most likely* to be appropriate?

- A) Rotate from relative value strategies into equity strategies. 
- B) Rotate from equity strategies into opportunistic strategies. 
- C) Rotate from equity strategies into distressed securities funds. 

## Explanation

The first objective is to increase the liquidity of the fund of hedge funds. This is unlikely to be achieved under rotating from equity strategies to distressed securities because equity strategies comprise long/short, short-bias, and market-neutral strategies, all of which are relatively liquid styles—whereas distressed securities funds are likely to be illiquid.

The second objective is to decrease the leverage of the fund. This is unlikely to be achieved under rotating from equity strategies into opportunistic strategies because opportunistic funds such as managed futures and global macro funds tend to use high levels of leverage. While equity market neutral funds tend to use high levels of leverage, long/short equity funds vary in their use of leverage, and short-bias funds use low levels of leverage. Hence, moving from equity strategies into opportunistic strategies is most likely to increase the leverage exposure of CAM.

The third objective is to lower exposure to significant credit events in the fund. Tail events, such as a credit crunch, can significantly negatively impact convertible arbitrage returns due to their long credit exposure and disrupt the spread strategies employed by fixed-income arbitrage funds. Both styles of fund are relative value funds; hence, allocating away from relative value funds is appropriate. The answer choice to rotate from relative value strategies into equity strategies meets this requirement. Note that this answer choice would also be expected to increase the liquidity of the portfolio because equity strategies are generally liquid, while relative value strategies have variable levels of liquidity. The answer choice also should lead to lower exposure to leverage in the portfolio because relative value funds tend to make extensive use of leverage.

Overall, the answer choice to rotate from relative value strategies into equity strategies is the best recommendation to meet the objectives of the investment committee.

(Module 34.3, LOS 34.e)

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## Question #38 - 39 of 49

Question ID: 1590516

Based on the fee structure information provided, if CAM allocates equally to two managers—one of which makes a gross return of 8%, while the other manager realizes a gross return of -8%—the net return to CAM investors is *closest* to:

A) 0.0%.



B) -1.5%.



C) -3.1%.



## Explanation

Management fee for the profitable manager = 1.5%

Incentive fee for the profitable manager =  $0.175 \times (8\% - 1.5\%) = 1.1375\%$

Net return from profitable manager =  $8\% - 1.5\% - 1.1375\% = 5.3625\%$

Management fee for the loss-making manager = 1.5%

Incentive fee for the loss-making manager = 0%

Net return from profitable manager =  $-8\% - 1.5\% = -9.5\%$

Hence, the gross return to CAM investors =  $(0.5 \times 5.3625\%) + (0.5 \times -9.5\%) = -2.06875\%$

Management fee payable by CAM investors = 1%

Incentive fee payable by CAM investors = 0%

So, the gross return earned by CAM investors =  $-2.06875\% - 1\% = -3.06875\%$

(Module 34.3, LOS 34.g)

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### Question #39 - 39 of 49

Question ID: 1590517

How many of the three factors listed by the client of CAM would a fund-of-funds (FoF) structure be preferred to a multi-manager structure from the point of view of investors in the fund?

A) Two.



B) None.



C) One.



#### Explanation

Only the operational risk factor is favorable for the FoF structure versus a multi-manager fund structure.

Fewer FoFs fail due to operational risk factors due to the diversification of the structure over multiple underlying funds with different operational risk exposures, and the lower level of leverage in FoFs versus multi-manager funds. Hence, from an operational risk perspective, FoFs are preferable to multi-manager funds.

Multi-manager funds are able to reallocate funds to different styles tactically faster than FoFs because multi-manager funds are not subject to the notice periods or lockups of external underlying funds. Hence, from a speed of tactical asset allocation perspective, the multi-manager structure is preferred.

Incentive fees are more likely to be able to be netted off across the performance of different teams in a single multi-manager structure rather than in an FoF, where external managers will take incentive fees on positive performance regardless of whether the overall FoF is profitable.

(Module 34.3, LOS 34.g)



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**Question #40 of 49**

Question ID: 1590497

A convertible bond arbitrage portfolio has begun implementing a strategy by purchasing convertible bonds with conversion prices substantially below the current stock price. The convertible bonds will:

- A)** behave most like a straight bond.
- B)** be impossible to hedge given the low delta.
- C)** behave most like the underlying equity.

**Explanation**

Since the convertible bond's current conversion price is well below the current stock price, the convertible bonds are substantially in the money and the convertible bonds will behave like the underlying equity.

In contrast, if the convertible bond's current conversion price is well above the current stock price, the convertible bonds are substantially out of the money and the convertible bonds will behave like straight bonds.

Even with a low delta, the delta risk (and gamma risk) can be hedged and is a typical strategy for a convertible arbitrage portfolio.

(Module 34.2, LOS 34.d)

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**Question #41 of 49**

Question ID: 1590489

Long/short equity funds typically:

- A)** eliminate market exposure through a net beta of zero.
- B)** reduce standard deviation to zero.
- C)** maintain a net long equity exposure of 40–60%.

**Explanation**

Unlike a market neutral approach, long/short approaches maintain a net long equity position typically between 40% and 60%. The net long equity position would have a positive beta. Managers aspire to have a standard deviation about half of a long-only approach.

(Module 34.2, LOS 34.b)

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### Question #42 of 49

Question ID: 1590493

Distressed security investing generally:

- A) involves shorting the securities of the distressed firm. 
- B) produces lower returns than other event-driven strategies. 
- C) uses low leverage. 

#### Explanation

Given the illiquidity and inherent risk of distressed securities, the strategy is typically employed with low leverage. Long investments are more common than shorting strategies, and average returns are generally high.

(Module 34.2, LOS 34.c)

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### Question #43 of 49

Question ID: 1590505

Top down analysis is the key driver behind which of the following hedge fund strategies?

- A) Fixed income arbitrage. 
- B) Merger and acquisition. 
- C) Global macro. 

#### Explanation

Global macro is focused on getting the big picture right on interest rate and equity market movements around the globe.

(Module 34.3, LOS 34.e)

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### Question #44 of 49

Question ID: 1590495

What are two liquidity issues that convertible arbitrage managers must successfully manage?

- A) Convert arbitrage is a relatively small investment strategy and traders compete with each other for attractive bonds and also for equity options to hedge with. 
- B) Investment grade issuance is highly cyclical and convertible bond managers are frequently unable to take positions in these bonds unless premiums are very low. 

- C) Issuance size tends to be small from relatively unproven companies and the shares to sell short may be difficult to locate and borrow.



### Explanation

Convertible bonds are typically smaller sized debt offerings from unrated companies. Frequently, shares of these companies are hard to borrow.

(Module 34.2, LOS 34.d)

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## Question #45 of 49

Question ID: 1590509

An advantage and a disadvantage of using fund-of-funds (FoF) for access to hedge fund are:

- A) The advantage of access to top ranked managers with smaller investment bite sizes and the disadvantage of no netting of performance fees.



- B) The advantage of lower overall fee structures and the disadvantage of a lack of research capability.



- C) The advantage of being able to access best of breed managers in each class of hedge fund and the disadvantage of negative economies of scale for monitoring.



### Explanation

Smaller family offices and high net worth investors benefit from FoFs as they enable access to top managers with smaller bite sizes and provide investment allocation guidance. Fees are substantially higher for FoFs, transparency generally lower, and there is no netting of performance fees, unlike multi-strategy funds. Netting of performance fees is an advantage since it may deny incentive payments to successful underlying funds when the overall performance of the FoF is poor. Therefore, not having netting is a disadvantage. Additionally, one of the advantages of FoFs is the economies of scale for monitoring.

(Module 34.3, LOS 34.g)

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Glen Downing, CFA, recently took a new position as chief investment officer at Deep Dive Asset Management (Deep Dive), a hedge fund-of-funds (FoF) based in the Cayman Islands. Having worked only at mutual funds in the past, Downing is new to the hedge fund world and wants to get up to speed quickly to take advantage of what he believes to be exceptional market opportunities. Downing is particularly excited about the opportunities an FoF can offer, compared to that of individual hedge funds.

The first fund Downing considers for investment is Copernicus Management (Copernicus), a hedge fund that invests in many different asset classes. Copernicus's team relies on a top-

down approach to screening potential investments, and the team's choice is often influenced by current market conditions. Downing will compare Copernicus to others in its peer group, but is unsure of how to classify the fund.

Another potential investment under consideration is APM International Advisors' global macro fund. APM management recently made the following claims during a sales pitch to Downing:

- **Claim 1:** "We tend to outperform other strategies in low-volatility markets."
- **Claim 2:** "Our strategies attempt to take advantage of markets that are not mean reverting."
- **Claim 3:** "Our high use of leverage—often in excess of 500% of capital—helps us to outperform other funds."

Finally, Downing is interested in a recent pitch made by Esoteric Investors (Esoteric), a hedge fund specializing in volatility trading. Downing has summarized Esoteric's pitch, breaking out its purported advantages into three parts:

- *Insurance-like returns.* Downing feels that adding Esoteric to his FoF should provide Deep Dive an exposure to insurance-like returns, because Esoteric often is a seller of VIX futures contracts.
- *Risk reduction.* Downing believes that adding an allocation to Esoteric will reduce overall portfolio risk due to the negative correlation between its long-volatility contracts and equity market returns.
- *Convexity.* Esoteric's salespeople told Downing that the fund's long-volatility positions exhibit strong negative convexity—another added benefit.

### Question #46 - 49 of 49

Question ID: 1590524

Downing's excitement about FoF investing would *least appropriately* be based on a belief that:

- A) he can provide better liquidity terms to his individual investors. 
- B) he may be able to invest in hedge funds closed to new investors. 
- C) his investors will have more transparency in their investments. 

### Explanation

FoF investments often provide more favorable redemption terms than do individual hedge funds. In addition, an FoF may be able to invest in closed hedge funds, due to previously arranged terms. One disadvantage, however, is that an FoF may not provide the same transparency of investments as individual funds do; this represents additional risk for investors.

(Module 34.3, LOS 34.g)

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### Question #47 - 49 of 49

Question ID: 1590525

Copernicus's investment strategy would *most accurately* be classified as:

- A) specialist. 
- B) opportunistic. 
- C) event driven. 

#### Explanation

Copernicus would be considered an opportunistic hedge fund because it invests across many asset classes, taking a macro-first, top-down view that considers which investments would be most favorable given current market conditions.

(Module 34.1, LOS 34.a)

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### Question #48 - 49 of 49

Question ID: 1590526

Which claim made by APM is *least accurate*?

- A) Claim 3. 
- B) Claim 2. 
- C) Claim 1. 

#### Explanation

Claim 1 is inaccurate: global macro funds are likely to *underperform* in low-volatility markets. Claim 2 is accurate; global macro funds also perform worse in mean-reverting markets. Claim 3 is plausible; global macro funds generally apply significant leverage to boost returns (often between 600%–700% of capital).

(Module 34.2, LOS 34.c)

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Esoteric's staff's statements regarding its volatility trading are *least accurate* regarding:

A) risk reduction.



B) convexity.



C) insurance-like returns.



**Explanation**

Some long volatility positions (e.g., long positions in variance swaps) exhibit *positive* (not negative) convexity, which indeed would be an advantage of adding this asset class. Volatility is highly negatively correlated with equity market returns, because spikes in volatility often coincide with sharp movements downward in equity markets. Thus, long-volatility contracts have a high negative correlation with equity. Selling volatility is a way to capture premia and provide regular cash flows, similar to insurance underwriting.

(Module 34.4, LOS 34.h)